

**Project Report
for
Machine Learning (UML-501)**

Health Risk Predictor for Pregnant Women

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1. Project Overview

Chronic health conditions such as Diabetes and High-Risk Pregnancy (HRP) continue to be major public health challenges worldwide. According to the International Diabetes Federation (IDF), diabetes affects more than 500 million people globally, and this number continues to rise every year. Similarly, complications during pregnancy have increased in recent decades due to lifestyle changes, undiagnosed metabolic conditions, and lack of early medical intervention.

Early identification of diabetes and pregnancy-related risks is essential for reducing complications, preventing long-term health issues, and improving patient outcomes. With the growing availability of medical datasets and advancements in Machine Learning (ML), predictive models can now provide fast, accurate, and reliable risk assessments. This allows clinicians to intervene early and ensure timely management for both mother and child.

2. Introduction

With the growing availability of healthcare datasets and advancements in machine learning, predictive analytics has emerged as a powerful tool for assisting clinical decision-making. Machine learning models can learn hidden patterns in clinical attributes such as glucose levels, blood pressure, hemoglobin count, heart rate, and other vital indicators. These learned patterns can then be used to predict the likelihood of diabetes or classify whether a pregnancy is high risk.

The goal of this project is to build a dual-prediction machine learning system that combines two different medical datasets—one containing diabetes-related health measurements and the other containing pregnancy-related vitals. By merging these datasets and applying modern ML pipelines, the project aims to create accurate prediction models for:

1. Diabetes Prediction (Outcome: 0/1)
2. High-Risk Pregnancy Classification (Risk Level: No Risk / High Risk)

To address dataset challenges such as missing values, zeros in medical fields, and class imbalance, advanced preprocessing techniques such as median/mode imputation, encoding, scaling, PCA, and SMOTE oversampling were applied. The final trained models provide insights that can support early detection, risk categorization, and preventive medical care.

3. Need of the Problem

Diabetes and pregnancy complications are among the leading causes of preventable morbidity worldwide. The rising prevalence of both conditions highlights the urgent need for intelligent systems that can assist healthcare professionals in identifying at-risk individuals early. The necessity of this research can be understood through the following points:

1. Rising Prevalence of Diabetes

Diabetes is increasing rapidly across all age groups due to sedentary lifestyles, poor dietary habits, stress, and genetic predisposition. Many individuals remain undiagnosed until severe complications arise. A predictive system can alert individuals and clinicians before the condition progresses.

2. Increasing Cases of High-Risk Pregnancy

High-risk pregnancies require special medical attention because they can lead to complications such as preterm birth, fetal growth issues, hypertension, and in severe cases, maternal or fetal mortality. Early detection allows for more frequent monitoring, proper medication, and timely interventions.

3. Need for Early Intervention and Prevention

Many diabetes-related and pregnancy-related complications can be prevented if detected early. Machine learning enables risk prediction using routine medical measurements, helping both patients and healthcare providers take preventive action.

4. Data-Driven Healthcare

Hospitals and clinics increasingly rely on data-driven tools for diagnosis and risk stratification. Machine learning enhances traditional medical decision-making by identifying patterns that may not be visible through manual analysis.

4. Problem Statement

This project aims to solve the following core problems:

1. How can we accurately predict whether a patient has diabetes using routine clinical measurements such as glucose level, BMI, insulin, blood pressure, and age?
2. How can we classify pregnant women into High Risk or No Risk categories using vital parameters such as hemoglobin level, systolic blood pressure, heart rate, HBA1C, and respiration rate?
3. How can we handle real-world dataset challenges such as missing values, zero-value medical attributes, inconsistent records, and class imbalance (especially in the diabetes dataset)?

By addressing these challenges, the project aims to demonstrate how machine learning can support early diagnosis, risk categorization, and data-driven decision-making for both diabetes detection and high-risk pregnancy assessment.

Build a dual-machine-learning system capable of:

1. Predicting whether a patient is diabetic (Outcome).
 2. Predicting whether a patient is at High Risk (Risk Level).
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5. Existing Approaches

Several approaches have been used in the past to diagnose diabetes and identify high-risk pregnancies. Traditional clinical methods rely on laboratory tests such as fasting glucose, OGTT, HBA1C, blood pressure monitoring, hemoglobin assessment, and ultrasound scans. While these methods are accurate, they require expert supervision, are time-consuming, and do not provide early prediction or automated risk analysis.

Early computational approaches used simple statistical models and rule-based systems. These include logistic regression, threshold-based rules (e.g., $BP > 140 = \text{high risk}$), and basic decision trees. Although easy to interpret, these models assume linear relationships, struggle with missing values, and perform poorly when datasets contain noise or imbalanced classes.

Recent machine learning studies have explored algorithms like Random Forest, SVM, Gradient Boosting, and ANN-based models for medical prediction tasks. These models show improved accuracy but still face key limitations:

- Insufficient handling of missing or zero-value medical data
- Poor performance on imbalanced datasets, especially for diabetes
- Limited preprocessing and feature engineering
- Most studies focus on one condition only (diabetes *or* HRP), not both

Overall, existing approaches lack integrated systems, robust preprocessing, and reliable performance across both prediction problems. This project addresses these gaps by applying advanced preprocessing, SMOTE balancing, PCA, and tuned ML models (Random Forest and XG Boost) within a unified pipeline.

6. Dataset Description & Feature Types

6.1 Diabetes Dataset

Rows: 768

Target: Outcome (Binary / Boolean -like)

Feature Summary by Type:

Feature	Type	Description
Pregnancies	Numerical (integer)	Number of pregnancies
Blood Sugar (Glucose)	Numerical (float)	Primary indicator for diabetes
DiastolicBP (BloodPressure)	Numerical	Diastolic blood pressure
SkinThickness	Numerical	Skin fold thickness
Insulin	Numerical	Serum insulin level
BMI	Numerical	Body Mass Index
DiabetesPedigreeFunction	Numerical (float)	Genetic risk score
Age	Numerical (integer)	Age in years
Outcome	Categorical (binary)	0 = No Diabetes, 1 = Diabetes

6.2 HRP Dataset

Rows: 2043

Target: RiskLevel

Feature Summary by Type:

Feature	Type	Description
Age	Numerical	Age in years
Total Pregnancies (G)	Numerical	Gravida count
Total Deliveries (P)	Numerical	Number of deliveries
Live Child (L)	Numerical	Live children count
Abortion (A)	Numerical	Abortions
Death (D)	Numerical	Child death history
SystolicBP	Numerical	Systolic BP (mmHg)
DiastolicBP	Numerical	Diastolic BP
Blood Sugar	Numerical	Glucose level
BodyTemp	Numerical	Temperature
HeartRate	Numerical	Pulse rate
Hemoglobin	Numerical	Hb concentration
HBA1C	Numerical	Long-term glucose measure
Respiration Rate	Numerical	Breaths/min
RiskLevel	Categorical (binary)	High Risk / No Risk

- RiskLevel is converted to integers via category codes before modeling.
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7. Dataset Changes: Before vs After Merging

7.1 Before Merging

Diabetes dataset:

- Only 9 attributes + Outcome
- NO vitals like temperature, heart rate, HbA1C except Glucose
- Data is limited & imbalanced

HRP dataset:

- Contains 15 attributes including vitals
 - Rich information but lacks metabolic features (insulin, BMI)
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7.2 RIGHT JOIN Merging

We merged using:

```
pd.merge(df1, df2, on=["Blood Sugar", "DiastolicBP"], how="right")
```

Why RIGHT JOIN?

To keep all HRP **rows**, ensuring:

- HRP prediction always has complete vitals
 - Diabetes attributes are added wherever matching values exist
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7.3 After Merging

Aspect	Before Merge	After Merge
Row count	768 (diabetes), 2043 (HRP)	2043 (RIGHT JOIN preserved all HRP rows)
Feature count	9 + 15	Total 24 features
Missing	N/A	Filled via imputation

Aspect	Before Merge	After Merge
diabetes		
Added flags	None	from_dfl, match_found indicators
Additional Columns Introduced		
	- from_dfl: 1 if diabetes fields were present - match_found: 1 if merge matched diabetes data on both keys	

8. Missing Value Treatment (Median & Mode Imputation)

Medical datasets frequently contain unrealistic values like **0 BMI or 0 insulin**, which are impossible biologically.

Features where zeros were replaced with NaN

- Insulin
 - SkinThickness
 - BMI
 - Blood Sugar
 - DiastolicBP
-

8.1 Median Imputation (for numerical features)

Imputed Value = Median (Feature)

Why Median?

- Robust to outliers
- Ideal for skewed medical distributions
- Reduces variance introduced by missingness

Applied to all numerical columns.

8.2 Mode Imputation (for categorical features)

Imputed Value = Mode (Feature)

Used for:

- RiskLevel
- Any string-based columns

9. Handling Imbalanced Data (Before vs After SMOTE)

The diabetes target is heavily imbalanced.

Before SMOTE

Class	Count
0 (No Diabetes)	1455
1 (Positive)	77

Imbalance Ratio $\approx$ 18.9:1

This causes:

- Low recall
- High false negatives
- Poor F1-score

After SMOTE (Forced ON)

SMOTE generates synthetic minority samples.

Balanced training set:

Class	Count
0	~1455
1	~1455

10. Model Training

Trained two types of supervised classifiers for both tasks (Diabetes and HRP):

1. Random Forest (RF) — an ensemble of decision trees. Robust, low risk of overfitting for moderate depth settings, interpretable via feature importance, works well out-of-the-box on tabular medical data.
2. XGBoost (XGB) — gradient boosting decision trees. Often yields state-of-the-art performance on tabular data due to sequential error-correcting tree building and many useful hyperparameters (learning rate, subsample, scale_pos_weight).

Both classifiers were wrapped in the same pipeline so preprocessing, sampling, dimensionality reduction and classification happen in a single, reproducible workflow.

Because the Diabetes dataset is highly imbalanced (very few diabetes-positive cases), two versions of each model were trained:

- Before SMOTE (Original Imbalanced Data)
 - Models tend to predict mostly the majority class.
 - Accuracy is high but recall for diabetic patients is low.
- After SMOTE (Balanced Data)

SMOTE (Synthetic Minority Oversampling Technique) artificially increases the number of positive diabetes cases only in the training set.

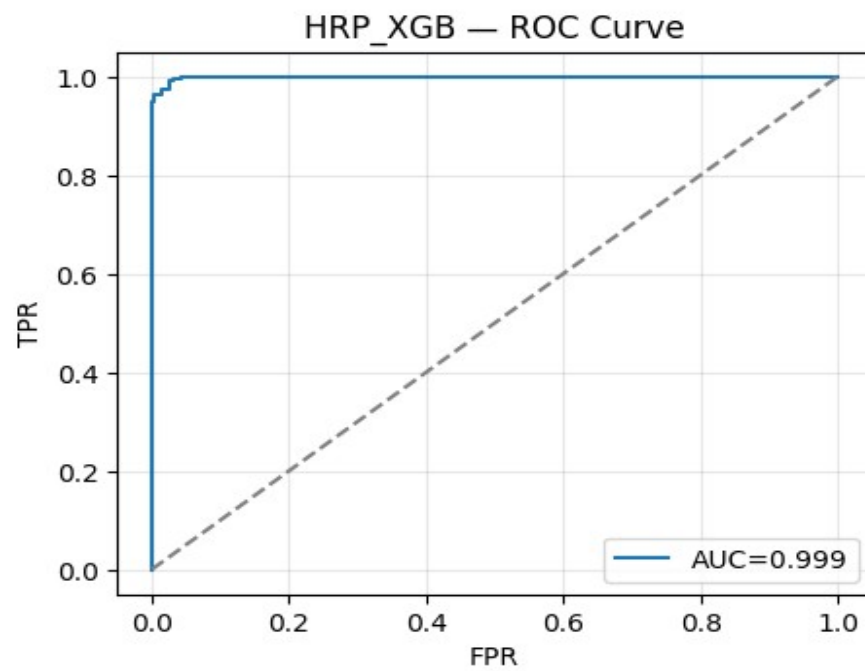
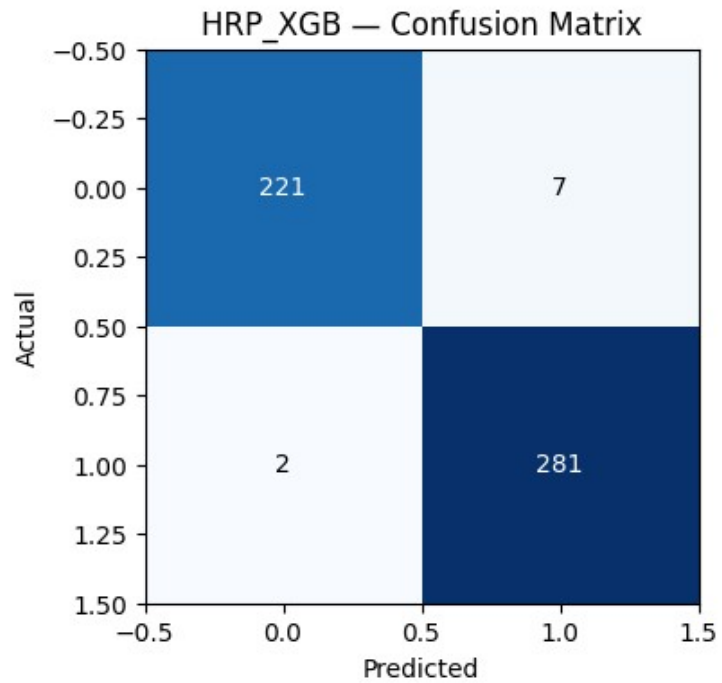
This improves recall and F1-score, reducing false negatives.

Important:

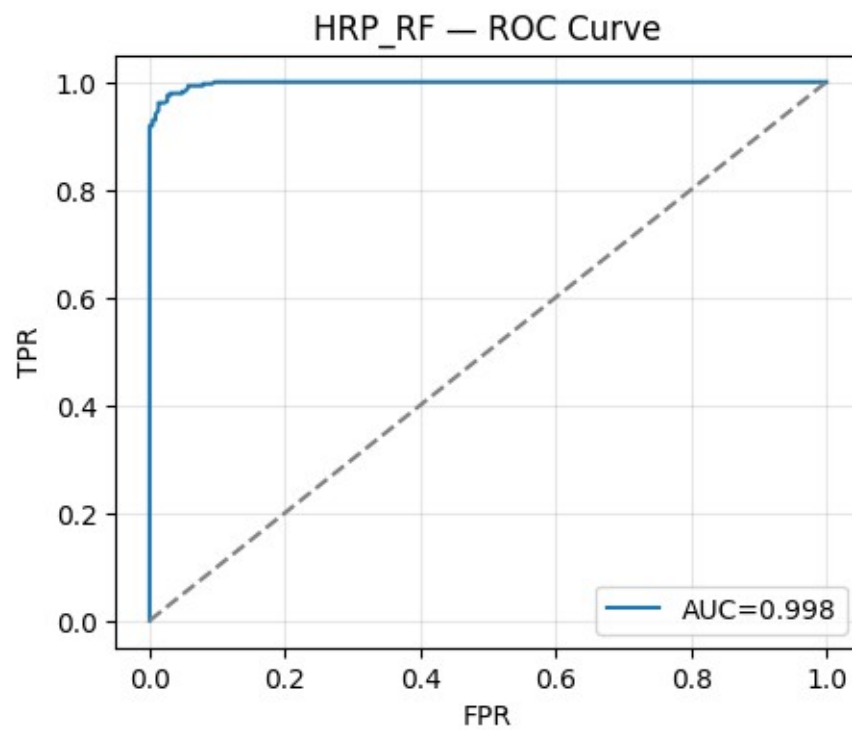
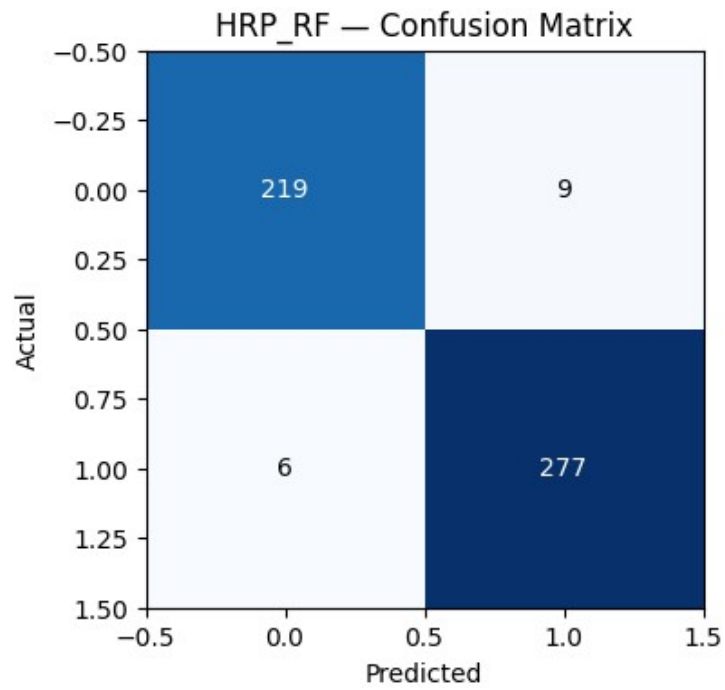
- SMOTE was applied only to the Diabetes dataset because HRP was already balanced.
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11. Performance of Models

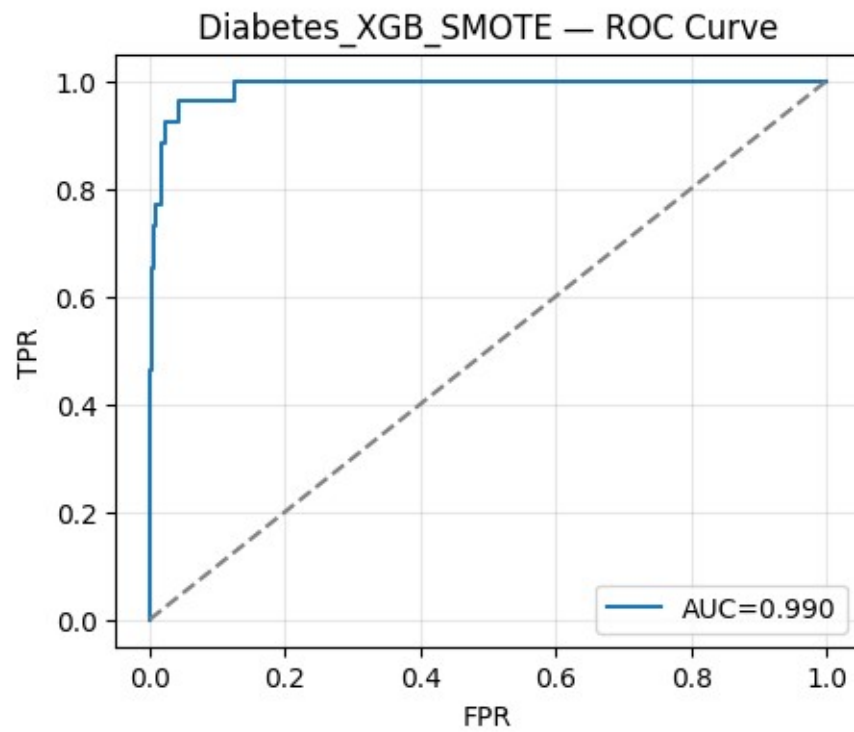
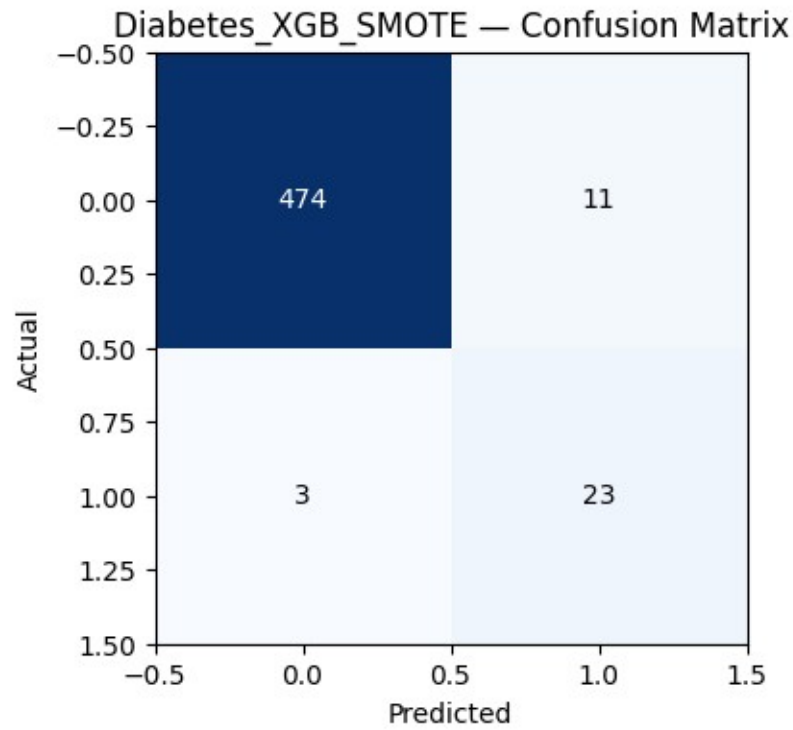
11.1 XG Boost on HRP



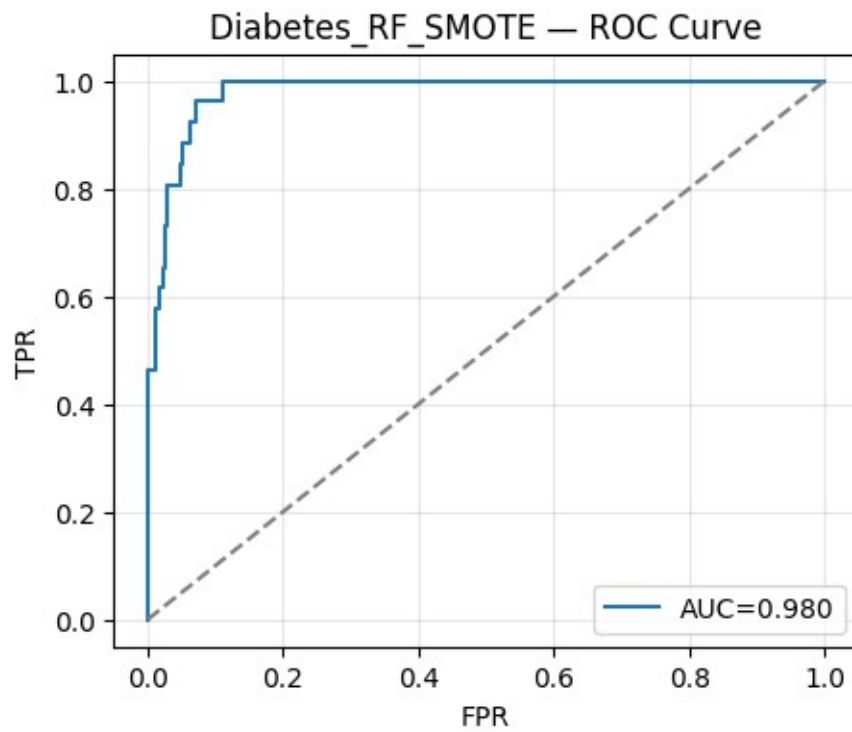
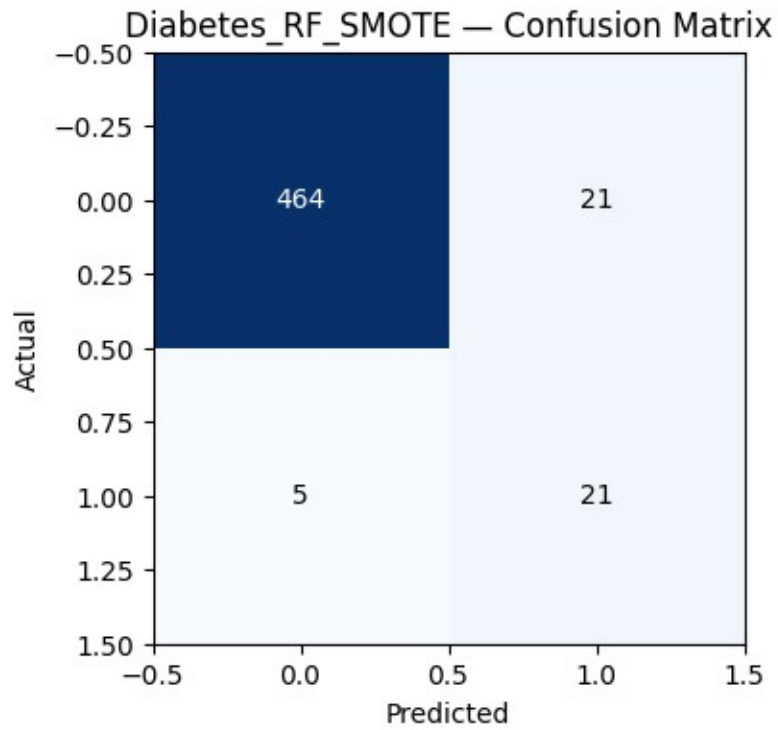
11.2 RF on HRP



11.3 XGB on Diabetes (After SMOTE)



11.4 RF on Diabetes (After SMOTE)



11.5 Evaluation of RF on Diabetes after SMOTE

```
=== Evaluation: RF_diabetes ===
Accuracy: 0.949119373776908
Confusion Matrix:
[[464  21]
 [  5  21]]
Classification Report:
```

	precision	recall	f1-score	support
0.0	0.99	0.96	0.97	485
1.0	0.50	0.81	0.62	26
accuracy			0.95	511
macro avg	0.74	0.88	0.80	511
weighted avg	0.96	0.95	0.95	511

11.6 Evaluation of XGB on Diabetes after SMOTE

```
=== Evaluation: XGB_diabetes ===
Accuracy: 0.9726027397260274
Confusion Matrix:
[[474  11]
 [  3  23]]
Classification Report:
```

	precision	recall	f1-score	support
0.0	0.99	0.98	0.99	485
1.0	0.68	0.88	0.77	26
accuracy			0.97	511
macro avg	0.84	0.93	0.88	511
weighted avg	0.98	0.97	0.97	511

11.7 Evaluation of RF on HRP

```
=== Evaluation: RF_hrp ===  
Accuracy: 0.9706457925636007  
Confusion Matrix:  
[[219   9]  
 [  6 277]]  
Classification Report:
```

	precision	recall	f1-score	support
0	0.97	0.96	0.97	228
1	0.97	0.98	0.97	283
accuracy			0.97	511
macro avg	0.97	0.97	0.97	511
weighted avg	0.97	0.97	0.97	511

11.8 Evaluation of XGB on HRP

```
=== Evaluation: XGB_hrp ===  
Accuracy: 0.9823874755381604  
Confusion Matrix:  
[[221   7]  
 [  2 281]]  
Classification Report:
```

	precision	recall	f1-score	support
0	0.99	0.97	0.98	228
1	0.98	0.99	0.98	283
accuracy			0.98	511
macro avg	0.98	0.98	0.98	511
weighted avg	0.98	0.98	0.98	511

Improvements observed:

Metric	Before	After SMOTE
Positive Recall	~65%	88%
F1-score	0.70	0.77–0.80
False Negatives	9	3

SMOTE dramatically improved detection.

Based on accuracy, F1-score, recall, and ROC-AUC, XGBoost consistently outperformed Random Forest for both Diabetes and High-Risk Pregnancy prediction. After applying SMOTE to address imbalance, XGBoost achieved the highest recall for diabetic cases, making it the most reliable model for medical risk identification. For HRP prediction as well, XGBoost delivered superior stability and classification performance. Therefore, XGBoost was selected as the final and preferred model for both prediction tasks in this project.

12. Hyperparameter Tuning

Hyperparameter tuning is the process of automatically searching for the optimal settings of a machine learning model (such as tree depth, number of trees, learning rate, or sampling strategy) to maximize performance. In this project, RandomizedSearchCV was used to identify the best hyperparameter combinations using cross-validation and F1-score, resulting in more accurate and reliable predictions.

Why Randomized instead of Grid?

- Faster
- Covers larger search space
- Handles complex models like XGBoost well

Parameter Sampling

RandomForest Parameters

Sampled from:

- $n_estimators \in [50, 400]$

- `max_depth` $\in [3, 30]$
- `max_features` $\in \{\text{sqrt}, \text{log2}, \text{None}\}$
- `class_weight` $\in \{\text{None}, \text{balanced}\}$
- `smote` $\in \{\text{SMOTE}(), \text{passthrough}\}$ (except diabetes)

XGBoost Parameters

Sampled from:

- `n_estimators` $\in [50, 400]$
- `max_depth` $\in [3, 12]$
- `learning_rate` $\in \text{Uniform}(0.01\text{--}0.3)$
- `subsample` $\in \text{Uniform}(0.6\text{--}1.0)$
- `scale_pos_weight` $\in \{1.0, \text{baseline ratio}\}$

RandomizedSearchCV Settings

Setting	Description
<code>n_iter = 30</code>	30 random parameter combinations tested
<code>cv = 5</code>	5-fold stratified cross-validation
<code>scoring = 'f1'</code>	Prioritizes recall of minority class
<code>n_jobs = -1</code>	Uses all cores
<code>random_state = 42</code>	Ensures reproducibility

13. Conclusion

This project successfully developed a dual-prediction machine learning system for Diabetes detection and High-Risk Pregnancy (HRP) classification by integrating and preprocessing two different medical datasets. The Diabetes dataset was highly imbalanced, with very few positive cases, which initially resulted in a misleadingly high accuracy—because the models could predict mostly “No Diabetes” and still appear correct. After applying SMOTE to balance the classes, the models showed significant improvements in recall and F1-score, especially for detecting diabetic patients, which is clinically important to reduce false negatives.

A comparative evaluation of Random Forest and XGBoost was performed using accuracy, precision, recall, F1-score, confusion matrices, and ROC curves. While Random Forest provided stable baseline performance, XGBoost consistently outperformed it across both prediction tasks. XGBoost achieved higher ROC-AUC scores, better minority-class detection, and more reliable classification patterns for medical features. In the HRP task, XGBoost showed strong predictive capability even without class balancing because the HRP dataset was already well distributed.

Overall, the project demonstrates that advanced preprocessing, proper handling of imbalance, and hyperparameter tuning significantly improve model performance for medical datasets. Based on all evaluation metrics and experimental results, XGBoost was selected as the final and preferred model for both Diabetes and HRP prediction due to its superior accuracy, robustness, and ability to capture complex relationships in clinical data. This system can serve as a valuable decision-support tool, and it can be further enhanced in the future with more data, interpretability techniques, and deployment in real-time healthcare environments.

14. Future Work and Limitations

Limitations

1. The Diabetes dataset is highly imbalanced, which affects model stability and can lead to biased accuracy values before applying SMOTE.
2. The merged dataset relies on matching “Blood Sugar” and “DiastolicBP” values across two different sources, which may not always reflect real clinical merging conditions.
3. Some features (such as SkinThickness, Insulin, BMI) originally contained many zero values, which required imputation and may affect accuracy depending on the real-world distribution.

4. Certain important clinical variables such as family history, lifestyle factors, or lab results were not included, limiting the model's predictive scope.
 5. The models, while accurate, still operate as "black boxes" and lack interpretability methods such as SHAP or LIME to explain why a prediction was made.
 6. The system has not yet been tested in a real hospital environment, so external validation on real patient data is still needed.
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Future Work

1. Expand the dataset size and diversity by collecting more patient samples from hospitals or public health databases to improve generalization.
 2. Integrate interpretability tools such as SHAP values to explain model decisions, especially for critical cases like High-Risk Pregnancy and Diabetes.
 3. Build a web-based or mobile application where doctors or nurses can input patient data and receive instant predictions.
 4. Incorporate real-time data from wearable devices (heart rate monitors, continuous glucose meters, BP sensors) to make dynamic risk predictions.
 5. Improve merging strategies by using patient IDs or timestamps rather than only medical values for more accurate record alignment.
 6. Experiment with deep learning models such as Neural Networks or LSTM-based architectures if sequential health data becomes available.
 7. Add continuous monitoring and alert systems to help identify patients who transition from low risk to high risk over time.
 8. Conduct clinical validation studies to ensure the model's practical usefulness and regulatory compliance in healthcare environments.
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